

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$4x + 5y = 100$$
$$5x + 4y = 62$$

If the system of equations above has solution (x, y) , what is the value of $x + y$?

- Answer
- A. 0
 - B. 9
 - C. 18
 - D. 38

Correct Answer: C

Rationale

Choice C is correct. Adding the given equations yields $9x + 9y = 162$. Dividing each side of the equation $9x + 9y = 162$ by 9 gives $x + y = 18$.

Choice A is incorrect and may result from incorrectly adding the equations. Choice B is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This value is equivalent to $y - x$.